

PhilFreela: Tools and Technologies Used

Software, libraries, services, and pretrained models used to build the system

1. Programming Languages

Language	Version	Used for
JavaScript (JSX)	–	Website logic and user interface components
HTML and CSS	HTML5, CSS3	Page structure and styling
Python	3.10.11	AI service for identity verification
SQL (PostgreSQL)	–	Database tables, security policies, and database functions

2. Frontend (Website)

Tool	Version	Used for
React	19.3.0	Building the user interface
Vite	8.2.2	Development server and production build tool
@vitejs/plugin-react	6.1.1	React support in Vite
@vitejs/plugin-basic-ssl	2.3.0	HTTPS during development, so phone cameras work
React Router	7.18.3	Navigation between pages
Supabase JS Client	2.116.0	Connects the website to the database, login, and file storage
MediaPipe Tasks Vision	1.0.1	Live face tracking in the browser during the face scan
qrcode.react	4.2.0	QR code that opens identity verification on a phone
Google Fonts	–	Typefaces: Plus Jakarta Sans, Syne, Audiowide

3. Backend and Database (Supabase)

Tool	Version	Used for
Supabase	Cloud	Backend platform hosting the database and services below
PostgreSQL	–	Main relational database
Supabase Auth	–	Sign-up, login, and user sessions
Supabase Storage	–	Uploaded files such as media and ID photos
Supabase Realtime	–	Live chat messages and announcements
Row Level Security and RPC functions	–	Data access rules and server-side database logic

4. AI Service (Identity Verification)

Tool	Version	Used for
FastAPI	0.141.1	Python web API for the AI service
Uvicorn	0.53.0	Server that runs the FastAPI app
Pydantic	2.13.5	Validates request and response data
DeepFace	0.0.101	Face detection and face recognition library
TensorFlow	2.21.0	Runs the DeepFace models
tf-keras	2.21.0	Keras support that DeepFace requires
OpenCV (opencv-python)	5.0.0	Image processing and the YuNet face detector
Pillow (PIL)	12.3.0	Opening, rotating, and resizing photos
NumPy	2.2.6	Numerical and image array operations
supabase-py	2.31.0	Saves verification results to the database
python-dotenv	1.2.3	Loads settings from the .env file
python-multipart	0.0.32	Receives photo uploads

5. Pretrained AI Models

Model	Runs on	Used for
MediaPipe Face Landmarker	Browser	Tracks the face live during the face scan
YuNet	AI service	Fast face detection on ID photos and selfies
RetinaFace	AI service	Backup face detection when YuNet finds no face
ArcFace	AI service	Face matching between the ID photo and the selfie

6. Development Tools

Tool	Version	Used for
Visual Studio Code	–	Code editor
Node.js	24.20.0	JavaScript runtime for the frontend tooling
npm	11.19.0	Installs frontend packages
Git	2.55.0	Version control
GitHub	–	Online code repository
Oxlint	1.82.0	Checks code quality
Playwright	1.63.0	Browser automation for testing pages
Supabase Dashboard and Supabase MCP	–	Database management
Claude Code (Anthropic)	–	AI coding assistant

Versions are the ones installed in the project as of September 25, 2026.